

COSC/MATH 4340: Numerical Methods for Differential Equations

Numerical Linear Algebra (NLA): Spectral Methods

Class: COSC/MATH 4340

Semester: Spring 2024

Instructor: Dane Taylor

Office: Eng 4087

Email: dane.taylor@uwyo.edu





Spectral Decompositions

Definition

Let A be a square, symmetric matrix of size n . λ_i and $\mathbf{v}_i$ are an *eigenvalue* and *eigenvector* of A if the following is true

$$A\mathbf{v}_i = \lambda_i\mathbf{v}_i$$

Definition

Each λ_i is a root of the *characteristic polynomial*

$$p(\lambda) = \det(A - \lambda I)$$

Note: Each $\mathbf{v}_i$ is in the *null space* of matrix $(A - \lambda_i I)$ and is unique up to normalization and a sign change.

(See notes for example.)



Eigenvalues and Eigenvectors

Definition

Let A be a square, symmetric matrix of size n . λ_i and $\mathbf{v}_i$ are an *eigenvalue* and *eigenvector* of A if the following is true

$$A\mathbf{v}_i = \lambda_i\mathbf{v}_i$$

Definition

Each λ_i is a root of the *characteristic polynomial*

$$p(\lambda) = \det(A - \lambda I)$$

Note: Each $\mathbf{v}_i$ is in the *null space* of matrix $(A - \lambda_i I)$ and is unique up to normalization and a sign change.

(See notes for example.)



Spectral Decompositions

Definition

Matrix A is *diagonalizable* if it has n eigenvalues (possibly repeated) and n linearly independent eigenvectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_N\}$. Define the matrices $D = \text{diag}[\lambda_1, \dots, \lambda_N]$ and $V = [\mathbf{v}_1, \dots, \mathbf{v}_N]$, then the *spectral decomposition* of A is given by

$$A = VDV^{-1} = \sum_j \lambda_j \mathbf{v}_j \mathbf{u}_j^T,$$

where $V^{-1} = U^T$ and $U = [\mathbf{u}_1, \dots, \mathbf{u}_N]$.

(See notes and code.)

Matrix Exponentials

Note: For any positive integer $k \geq 0$ one has

$$A^k = VD^kV^{-1} = \sum_j \lambda_j^k \mathbf{v}_j \mathbf{u}_j^T,$$

Definition

If matrix A is diagonalizable, then its *matrix exponential* is given by

$$\begin{aligned} e^A &= I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots \\ &= V(I + D + \frac{D^2}{2!} + \frac{D^3}{3!} + \dots)V^{-1} \\ &= Ve^D V^{-1} \\ &= \sum_j e^{\lambda_j} \mathbf{v}_j \mathbf{u}_j^T. \end{aligned}$$



Spectral Solution to Linear Systems of ODEs

Theorem

Consider the linear system of ODEs

$$\frac{d}{dt}\mathbf{x} = A\mathbf{x}$$

with initial condition $\mathbf{x}(0)$. If A is a square diagonalizable matrix, then the solution $\mathbf{x}(t)$ is given by

$$\mathbf{x}(t) = e^{tA}\mathbf{x}(0) = Ve^{tD}V^{-1}\mathbf{x}(0) = \sum_j c_j e^{t\lambda_j} \mathbf{v}_j,$$

where $c_j = [V^{-1}\mathbf{x}(0)]_j = [U^T\mathbf{x}(0)]_j = \mathbf{u}_j^T\mathbf{x}(0)$.

(See notes and code.)



Equilibrium and Stability of Systems of ODEs

Definition

If $\frac{d}{dt}\mathbf{x} = F(\mathbf{x})$ is a system of ODEs, then vector $\mathbf{x}^*$ is an *equilibrium* iff $F(\mathbf{x}^*) = \mathbf{0}$.

Lemma

For a linear system of ODEs, $F(\mathbf{x}) = A\mathbf{x}$, and $A\mathbf{x} = \mathbf{0}$ iff $\mathbf{x} \in \text{Null}(A)$. If A is diagonalizable, then $\text{Null}(A) = \mathbf{0}$ and so $\mathbf{x}^ = \mathbf{0}$ is the only equilibrium.*

Equilibrium and Stability of Systems of ODEs

Definition

An equilibrium of an ODE is called *asymptotically stable* if there exists a region $\mathcal{S} \in \mathbb{R}^n$ containing $\mathbf{x}^*$ such that $\mathbf{x}(t) \rightarrow \mathbf{x}^*$ for any initial condition $\mathbf{x}(0)$ within $\mathcal{S}$.

Definition

An equilibrium is *globally* asymptotically stable if $\mathbf{x}(t) \rightarrow \mathbf{x}^*$ for any $\mathbf{x}(0) \in \mathbb{R}^n$. It's *locally* asymptotically stable if $\mathbf{x}(t) \rightarrow \mathbf{x}^*$ only for a strict subset $\mathcal{S} \subset \mathbb{R}^n$.

Definition

An equilibrium is *unstable* if there exists initial conditions arbitrarily close to $\mathbf{x}^*$ that diverge from $\mathbf{x}^*$.



Equilibrium and Stability of Systems of ODEs

Lemma

The equilibrium $\mathbf{x}^ = \mathbf{0}$ of a linear system of ODEs, $\frac{d}{dt}\mathbf{x} = A\mathbf{x}$, is asymptotically stable if and only if all eigenvalues λ_i of A are negative.*

Lemma

The equilibrium $\mathbf{x}^ = \mathbf{0}$ is unstable stable if and only if one (or more) eigenvalue is positive.*

Lemma

If A has both positive and negative eigenvalues, then $\mathbf{x}^ = \mathbf{0}$ is called a saddle point.*



Robustness of Eigenvalues and Eigenvectors

Given the importance of matrix spectral properties for solving ODEs, it's important to

- ▶ understand their robustness to perturbations/error
- ▶ develop efficient methods to approximately find them.

Theorem (Bauer-Fike)

Let A be a diagonalizable matrix, $A = VDV^{-1}$. Let E be a perturbation to A , and let λ be an eigenvalue of matrix $A + E$. Then

$$\min_i |\lambda - \lambda_i| \leq \text{cond}(V) \|E\|.$$

(See notes for proof.)

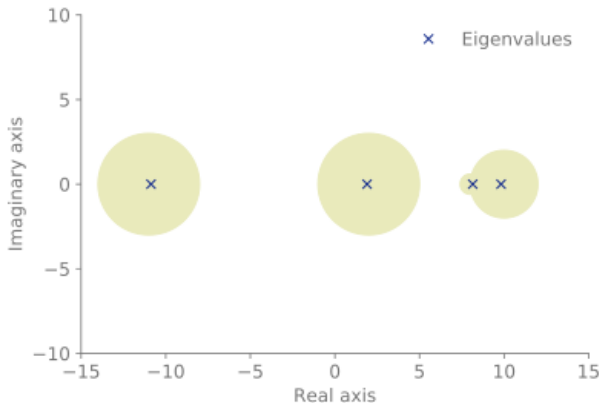


Robustness of Eigenvalues and Eigenvectors

Theorem (Gerschgorin Circles)

Let A be square matrix with diagonal entries $\{A_{ii}\}$ and for each i define a circle $\mathcal{C}_i = \{z; |z - A_{ii}| < r_i\}$ in the complex plane $\mathbb{C}$, where radius $r_i = \sum_{j \neq i} |A_{ij}|$ is determined by the size of the off-diagonal entries. Then each eigenvalue λ_i of A belongs to one of the circles, and if m circles are overlapping then exactly m eigenvalues are found in their union.

Robustness of Eigenvalues and Eigenvectors



(See notes for example.)



Robustness of Eigenvalues and Eigenvectors

Theorem (Matrix Spectral Perturbations)

Let $A(\epsilon) = A + \epsilon B$ be a diagonalizable matrix, $A = VDV^{-1}$, where $D = \text{diag}[\lambda_1, \dots, \lambda_N]$, $V = [\mathbf{v}_1, \dots, \mathbf{v}_N]$ and $V^{-1} = U^T$ with $U = [\mathbf{u}_1, \dots, \mathbf{u}_N]$. Assume the eigenvalues are non-repeated. Then one can approximate the eigenvalues and eigenvectors of $A(\epsilon)$ by

$$\lambda_k(\epsilon) = \lambda_k + \epsilon \frac{\mathbf{u}_k^T B \mathbf{v}_k}{\mathbf{u}_k^T \mathbf{v}_k} + \mathcal{O}(\epsilon^2)$$

$$\mathbf{v}_k(\epsilon) = \mathbf{v}_k + \epsilon a_k \mathbf{u}_k + \epsilon \sum_{j \neq k} \frac{\mathbf{u}_j^T B \mathbf{v}_k}{(\lambda_k - \lambda_j) \mathbf{u}_j^T \mathbf{v}_j} \mathbf{u}_j + \mathcal{O}(\epsilon^2)$$

(See notes for sketch of proof.)



Numerical Approximations for Spectral Properties

We now study different numerical techniques to approximate the eigenvalues and eigenvectors of matrices.

There are 2 main families of methods

- ▶ Iterative methods to approximate one or a couple eigenvalues and eigenvectors.
- ▶ Methods to compute ALL eigenvalues and eigenvectors.



Power Method

One can estimate the largest-in-magnitude eigenvalue λ_1 and it's eigenvector $\mathbf{v}^{(1)}$ via the following.

Definition

Let $\mathbf{x}^{(0)}$ be an initial guess at the eigenvector. The *power method*, or power iteration, is defined by the following iteration:

$$\begin{aligned}\mathbf{z}^{(m)} &= \mathbf{x}^{(m)} / \|\mathbf{x}^{(m)}\|_2 \\ \mathbf{x}^{(m+1)} &= A\mathbf{z}^{(m)} \\ \beta_m &= [\mathbf{z}^{(m)}]^T A\mathbf{z}^{(m)}\end{aligned}$$

Then $\mathbf{z}^{(m)} \rightarrow \mathbf{v}^{(1)}$ and $\beta_m \rightarrow \lambda_1$ as $m \rightarrow \infty$.



Power Method

Theorem

Consider a diagonalizable matrix A with largest-in-magnitude eigenvalue λ_1 and eigenvector $\mathbf{v}^{(1)}$, and assume $|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \dots$. Then the power method converges as

$$\|\mathbf{z}^{(m)} - \mathbf{v}^{(1)}\|_{\infty} \leq c \left| \frac{\lambda_2}{\lambda_1} \right|^m$$

for some constant $c > 0$, provided that the initial condition is not orthogonal to $\mathbf{v}^{(1)}$.

(See notes for proof.)



Power Method

Definition

The *Rayleigh-Ritz Quotient* is defined as

$$\mathcal{R}_A(\mathbf{z}) = \frac{\mathbf{z}^T A \mathbf{z}}{\mathbf{z}^T \mathbf{z}}$$

Theorem

The *Courant-Fischer Theorem* states that the largest eigenvalue is as

$$\lambda_{\max} = \max_{\mathbf{z}} \mathcal{R}_A(\mathbf{z}) = \max_{\mathbf{z}} \frac{\mathbf{z}^T A \mathbf{z}}{\mathbf{z}^T \mathbf{z}}.$$

For the power method, observe that $\beta_m = \mathcal{R}_A(\mathbf{z}^{(m)})$.



Inverse Iteration

Consider if one has an approximate eigenvalue $\lambda \approx \lambda_i$. Inverse iteration modifies the power method to estimate λ_i and its eigenvector $\mathbf{v}^{(i)}$.

Definition

Let $\mathbf{x}^{(0)}$ be an initial guess at eigenvector $\mathbf{v}^{(i)}$. *Inverse iteration* is defined as:

$$\begin{aligned}\mathbf{z}^{(m)} &= \mathbf{x}^{(m)} / \|\mathbf{x}^{(m)}\|_2 \\ \mathbf{x}^{(m+1)} &= (A - \lambda I)^{-1} \mathbf{z}^{(m)} \\ \beta_m &= [\mathbf{z}^{(m)}]^T A \mathbf{z}^{(m)}\end{aligned}$$

Then $\mathbf{z}^{(m)} \rightarrow \mathbf{v}^{(i)}$ and $\beta_m \rightarrow (\lambda_i - \lambda)^{-1}$ as $m \rightarrow \infty$.

(See notes.)



Inverse Iteration

Theorem

Let λ_i and λ_j be the closest and second-closest eigenvalues to λ . Then inverse iteration converges as

$$\|\mathbf{z}^{(m)} - \mathbf{v}^{(i)}\|_{\infty} \leq c \left(\frac{|\lambda_i - \lambda|}{|\lambda_j - \lambda|} \right)^m$$

for some constant $c > 0$, provided that the initial condition is not orthogonal to $\mathbf{v}^{(1)}$.

(See notes for proof.)



Notes on Power Method and Inverse Iteration

- ▶ For inverse iteration, one does not necessarily need to compute the inverse matrix $B = (A - \lambda I)^{-1}$. It may be faster to instead solve a linear system for $\mathbf{x}^{(m+1)}$:

$$(A - \lambda I)\mathbf{x}^{(m+1)} = \mathbf{z}^{(m)}$$

- ▶ Both the power method and inverse iteration can be implemented efficiently for large sparse matrices using sparse matrix-vector multiplication

Rayleigh Quotient Iteration (RQI)

Can we replace λ for inverse iteration by our estimate β_m ? Yes.

Definition

Rayleigh Quotient Iteration is defined as:

$$\mathbf{z}^{(m)} = \mathbf{x}^{(m)} / \|\mathbf{x}^{(m)}\|_2$$

$$\beta_m = [\mathbf{z}^{(m)}]^T A \mathbf{z}^{(m)}$$

$$\mathbf{x}^{(m+1)} = (A - \beta_m I)^{-1} \mathbf{z}^{(m)}$$

Then $\mathbf{z}^{(m)} \rightarrow \mathbf{v}^{(i)}$ and $\beta_m \rightarrow \lambda_i$ as $m \rightarrow \infty$.

Note: For non-repeated eigenvalues, RQI converges quadratically!

(See code.)



Deflation

How can we find multiple eigenvalues and eigenvectors?

Definition

Let A be a square, symmetric matrix with an eigenvalue λ with right eigenvector $\mathbf{v}$ and left eigenvector $\mathbf{u}$. Then the *deflated* matrix is given by

$$A' = A - \lambda \mathbf{v} \mathbf{u}^T.$$

- ▶ Matrix A' still has a right eigenvector $\mathbf{v}$ and left eigenvector $\mathbf{u}$, but the associated eigenvalue is now: $\lambda - \lambda = 0$.
- ▶ When applied to A' , the power method returns the second-largest eigenvalue of A and its eigenvector. Deflating k times allows us to find the k top eigenvalues and eigenvectors.

(See code for example.)



QR Method to Compute all Eigenvalues

One can simultaneously compute all eigenvalues $\{\lambda_i\}$ of a diagonalizable matrix A via the QR iteration.

Definition

The *QR decomposition* of a matrix A is given by $A = QR$, where R is an upper triangular matrix and Q is a unitary matrix such that $Q^T Q = I$.

Definition

The *QR iteration* for an initial matrix A_0 is defined by

$$Q_m R_m = A_m$$

$$A_{m+1} = R_m Q_m$$

QR Method to Compute all Eigenvalues

Theorem

The sequence of matrices A_m for the QR iteration converges to a diagonal matrix D that contains the eigenvalues on the diagonal. Convergence satisfies the following bound

$$\|D - A_m\| \leq c \max_i \left| \frac{\lambda_{i+1}}{\lambda_i} \right|^m,$$

where we assume $|\lambda_1| > |\lambda_2| > \dots > |\lambda_n|$.

- ▶ The diagonal entries in A_m converge to the eigenvalues.
- ▶ For symmetric A , the product $Q_1 Q_2 \dots Q_m$ converges to the matrix V of eigenvectors.

(See notes and code.)



Householder Matrices and the QR Decomposition

But how does one construct the $A = QR$ decomposition?

One approach involves Householder matrices.

Definition

Given a vector $\mathbf{w}$ with $\|\mathbf{w}\|_2 = 1$, it's associated *Householder matrix* takes the form: $U = I - 2\mathbf{w}\mathbf{w}^T$

Theorem

U is symmetric and unitary.

(See notes for example and proof.)



Householder Matrices and the QR Decomposition

Householder matrices can be used to convert a column in A into a new column whose lower entries contain zeros.

Example: $A = \begin{bmatrix} 4 & 3 \\ 3 & -3 \end{bmatrix}$

(See notes and code.)